Estimating Persistent Explanatory Variables: A Machine Learning Approach

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Paper: SSRN 4636846 Code and example data

Persistent regressors and firm fixed effects

- Firm fixed effects control for unobserved, time-invariant firm characteristics.
- Many economic variables change slowly within firms: R&D, ownership, organizational practices, and managerial characteristics.
- FE estimates rely on the variation remaining within firms. That variation can be small and noisy.
- The resulting trade-off is between controlling for persistent heterogeneity and retaining information for precise estimation.

Question: Can we select informative firm indicators rather than include all of them?

Current manuscript, Introduction and Section 2. Persistence alone does not imply FE inconsistency.

Two empirical applications

R&D input and innovation output

- Innovation investment is persistent; patent production differs substantially across firms.
- Linear patent coefficients: OLS 0.593, FE 0.041, PRL 0.199, and DML 0.213.

Managerial ownership and firm value

- Ownership reflects persistent contracting environments and changes slowly within firms.
- OLS, FD, PRL, and DML yield a positive ownership coefficient and a negative squared term; both FE terms are individually insignificant.

Current manuscript, Tables 2 and 5; Himmelberg, Hubbard, and Palia (1999); Zhou (2001).

What changes when we include firm effects?

$$z_{it} = \eta_i + \nu_{it}, \qquad \widetilde{z}_{it} = z_{it} - \bar{z}_i = \nu_{it} - \bar{\nu}_i.$$

- Demeaning removes the permanent firm component η_i.
- A larger between-firm share leaves less within-firm variation.
- Over a short observation window, stronger persistence can further reduce variation after demeaning.
- Under strict exogeneity and identification, FE can remain consistent while estimates become imprecise.
- Measurement error is a separate problem: removing the true signal can increase the relative importance of remaining noise.

Current manuscript, Sections 2 and 4. The decomposition above is illustrative.

Our approach

1. Selective estimation
   Apply PRL and DML to firm indicators in linear and Poisson models.

2. Simulation evidence
   Vary the between-firm share, serial persistence, prevalence, and shared firm components.

3. Diagnostic guidance
   Describe available variation and map error/power targets to diagnostics under an assumed DGP.

4. Two applications
   Reexamine the R&D–innovation and ownership–firm value relations.

Adjusted Hausman–Taylor is a supplementary comparison, developed in Section 6.

Panel regression framework

$$y_{i,t+1} = \beta_0 + \gamma z_{it} + \mathbf{x}_{it}^{\top}\boldsymbol{\beta} + \alpha_i + \varepsilon_{it}.$$

- z_{it}: persistent explanatory variable of interest.
- $\mathbf{x}_{it}$: observed controls, including year indicators.
- α_i: time-invariant firm heterogeneity.
- γ: coefficient of interest.
- Empirical inference uses standard errors clustered by firm.

In the ownership application, ownership and its square enter jointly.

Current manuscript, Section 2.1. Empirical outcomes and controls follow the paper's timing conventions.

Conventional estimators

Estimator	Variation used	Main consideration
Pooled OLS	Between and within firms	Persistent omitted firm factors can confound the coefficient.
Firm FE	Deviations from firm means	Controls unrestricted firm intercepts; needs usable within variation.
FD	Consecutive-year changes	Removes firm intercepts; small annual changes can be especially noisy.

- FE and FD remove the same time-invariant firm component.
- Their relative efficiency also depends on the error process.
- Comparing coefficients requires attention to assumptions, samples, and the scale of the regressor change.

Current manuscript, Section 2.1; Wooldridge (2002); Breuer and deHaan (2024).

R&D and innovation data

- U.S. public firms from CRSP/Compustat; patent and citation information from PatentsView.
- Exclude financial and utility firms; require nonmissing, nonnegative total assets.
- Full sample: 86,341 firm-years and 11,544 firms, 1975–2000.
- Patenting-firm subsample: 45,913 firm-years and 4,312 firms.
- Explanatory variables precede the innovation outcome by one year.

Current manuscript, Section 5.2.1. FE and FD use smaller estimation samples, reported on the result slides.

R&D input and innovation outcomes

Primary R&D measure

$$\mathrm{R\&D/Assets}_{it} = \frac{\text{five-year accumulated R\&D expense}}{\text{total assets at fiscal year-end } t}.$$

- Linear outcomes: ln(1 + Patents), ln(1 + Citations), and ln(1 + Adjusted citations).
- Poisson outcomes: patent and citation counts.
- Controls include size, age, capital intensity, missing R&D, Tobin's *Q*, profitability, leverage, cash, financing constraints, institutional ownership, and competition.
- Use the paper's winsorized variables and full control set.

Current manuscript, Section 5.2.1 and Tables 2 and 4. Alternative R&D measures include R&D/ME and ln(1 + R&D).

R&D variation and persistence

Diagnostic	R&D/Assets
D1: between-firm variance share	0.801
D2: median within-firm autocorrelation	0.851
D2: interquartile range	0.625–0.950
D3: absorption by firm and year effects	0.806
Patent model: between correlation R^2	0.310
Patent model: within correlation R^2	0.020
Between/within correlation-R^2 ratio	15.207

Most variation is between firms, and R&D is highly persistent within firms. These statistics describe the available signal; they do not establish exogeneity.

Current manuscript, Table 1, full pooled sample. The ratio uses saved precision, not the displayed rounded R^2 values.

R&D and patents: linear estimates

Dependent variable: $\ln(1 + \text{Patents})$

	OLS	FE	FD	PRL	DML	adjHT
R&D/Assets	.593***	.041	−.031	.199***	.213***	.220***
	(.042)	(.028)	(.023)	(.018)	(.014)	(.027)
Selected firms				1,215	1,711	
Firm selection				10.52%	14.82%	
Observations	86,341	84,620	73,319	86,341	86,341	86,341

- PRL and DML retain positive, significant associations.
- FE and FD coefficients are close to zero and statistically insignificant.
- Coefficient differences alone cannot identify their source.

Current manuscript, Table 2, Panel A. Full controls; firm-clustered SE in parentheses. FE removes 1,721 singleton observations. FD uses consecutive-year differences. * $p < .10$, ** $p < .05$, *** $p < .01$.

Citations and adjusted citations

	OLS	FE	FD	PRL	DML	adjHT
ln(1 + Citations)						
R&D/Assets	1.396***	−.051	−.176**	1.397***	1.364***	.551***
	(.084)	(.068)	(.080)	(.083)	(.050)	(.065)
ln(1 + Adjusted citations)						
R&D/Assets	.590***	.033	−.051*	.210***	.198***	.231***
	(.045)	(.031)	(.031)	(.019)	(.015)	(.030)
Observations	86,341	84,620	73,319	86,341	86,341	86,341

The positive relation under pooled and selective estimation also appears for citation-based outcomes. FE coefficients remain insignificant; FD estimates are negative.

Current manuscript, Table 2, Panels B and C. Full controls and firm-clustered SE. Stars use the same thresholds as the preceding slide.

Selective firm effects

$$y_{i,t+1} = \beta_0 + \gamma z_{it} + \mathbf{x}_{it}^{\top}\beta + \sum_{s \in \mathcal{S}} \alpha_s \mathrm{dummy}_{s,i} + \varepsilon_{it}.$$

- Pooled estimation includes no firm indicators.
- Full FE allows unrestricted firm intercepts, subject to a normalization.
- PRL and DML select a subset of candidate firm indicators.
- In the empirical applications, year indicators are also candidates for selection; the reported controls enter as variables of interest.

Current manuscript, Section 2.2. Selective estimation requires its own identification and nuisance-approximation conditions.

Economic intuition for selective indicators

- After observed characteristics and common time effects are controlled for, some firms may retain large persistent deviations.
- Distinctive technologies, organizational practices, or governance arrangements can motivate such departures.
- A sparse approximation represents the firm component with a common baseline and a limited set of substantial deviations.
- Selection is a statistical adjustment. It does not identify a firm's culture or prove that unselected firm effects are exactly zero.

Current manuscript, Introduction, Section 2, and Internet Appendix concentration analysis. The selected share is sample- and specification-dependent.

Post-regularization LASSO (PRL)

1. Outcome equation
   Select firm/year indicators by LASSO; refit on the selected indicators and obtain outcome residuals.

2. Regressor equations
   Repeat selection and post-LASSO fitting for R&D and each included control, obtaining residualized regressors.

3. Target coefficient
   Regress the residualized outcome on the residualized regressors; use firm-clustered inference.

The main linear application uses BIC for LASSO tuning.

Current manuscript, Section 2.2; Chernozhukov, Hansen, and Spindler (2015). Implementation: `poregress`.

Double machine learning (DML)

1. Split firm-year observations into five folds.
2. Fit the nuisance models, including selection and post-LASSO refitting, on four training folds.
3. Use those fitted models to obtain residuals for the held-out fold.
4. Repeat until every observation has out-of-fold residuals.
5. Pool those residuals to estimate the target coefficient.

Cross-fitting separates nuisance fitting from evaluation.

Each training fit uses fewer observations, but the final stage uses residuals from all folds.

Current manuscript, Section 2.3; Chernozhukov et al. (2018). Implementation: `xporegress`, five folds.

Inference and assumptions

- Orthogonalization reduces sensitivity to nuisance-estimation errors.
- Cross-fitting helps control the dependence created by fitting nuisance functions on the same observations.
- Valid inference still requires exogeneity, sufficient identifying variation, and suitable approximation and rate conditions.
- Selective indicators do not automatically resolve simultaneity, measurement error, or omitted time-varying confounding.

PRL and DML provide additional specifications to evaluate alongside pooled, FE, and FD estimates.

Current manuscript, Section 2 and Internet Appendix theory. Descriptive concentration is not a formal test of the required rate conditions.

Count-based outcomes

$$\mu_{it} = \exp\left(\beta_0 + \gamma z_{it} + \mathbf{x}_{it}^{\top}\beta + \sum_{s\in\mathcal{S}} \alpha_s \mathrm{dummy}_{s,i}\right).$$

Here μ_{it} is the conditional mean of the count outcome.

- Poisson and FE Poisson use no firm indicators and the full set, respectively.
- PRL Poisson selects indicators using Poisson LASSO and weighted linear auxiliary regressions.
- The target parameters solve residual-based moment equations.
- DML Poisson adds cross-fitting.

Current manuscript, Section 2.4; Belloni, Chernozhukov, and Wei (2016). Patent and citation counts; adjusted citations are excluded.

R&D and innovation: Poisson estimates

	Poisson	FE Poisson	PRL Poisson	DML Poisson
Patent counts				
R&D/Assets	2.305***	−.248	2.407***	2.312***
	(.187)	(.255)	(.161)	(.120)
Firm selection			10.33%	21.29%
Citation counts				
R&D/Assets	2.094***	−.125	2.299***	2.262***
	(.219)	(.252)	(.256)	(.113)
Firm selection			10.32%	20.79%
Observations	86,341	84,620	86,341	86,341

Pooled and selective estimates imply similar positive associations. The full-FE estimates are insignificant.

Current manuscript, Table 4. Full controls; firm-clustered SE. Firm selection percentages exclude year indicators. *** $p < .01$.

Selection varies across outcomes and applications

Selected firm indicators	PRL	DML
Linear patents	1,215 (10.52%)	1,711 (14.82%)
Linear citations	500 (4.33%)	922 (7.99%)
Linear adjusted citations	1,168 (10.12%)	1,856 (16.08%)
Poisson patents	1,192 (10.33%)	2,458 (21.29%)
Poisson citations	1,191 (10.32%)	2,400 (20.79%)
Managerial ownership	580 (26.88%)	725 (33.60%)

- Candidate firms: 11,544 in R&D; 2,158 in ownership.
- There is no universal “10–20%” selected share.
- Selected indicators may capture outcome or regressor nuisance variation.

Current manuscript, Tables 2, 4, and 5. Counts refer to firms only; year indicators are reported separately in the paper.

Managerial ownership and firm value

- ExecuComp and Compustat: 23,879 firm-years and 2,158 firms.
- Source years: 2006–2024. Lagged-variable level-estimation sample: 2007–2024.
- Ownership aggregates all named executives' directly held common shares and unexercised exercisable options, divided by shares outstanding.
- Dependent variable: Tobin's Q, constructed following Himmelberg et al. (1999).
- Ownership, ownership squared, and controls enter with a one-year lag.

Current manuscript, Section 5.3.1 and Table 5. Ownership and its square are not winsorized.

Ownership variation and persistence

Diagnostic	Ownership
D1: between-firm variance share	0.700
D2: median within-firm autocorrelation	0.694
D2: interquartile range	0.382–0.866
D3: absorption by firm and year effects	0.726
Tobin's Q model: between correlation R^2	0.230
Tobin's Q model: within correlation R^2	0.051
Between/within correlation-R^2 ratio	4.469

Ownership is persistent, with much of its variation between firms. This reproduces the central information trade-off in a different economic setting.

Current manuscript, Table 1. Diagnostics use the pooled application sample. Ratios use saved precision.

Ownership and firm value: estimates

Dependent variable: Tobin's Q

	OLS	FE	FD	PRL	DML	adjHT
Ownership	1.945^{***}	$-.312$	1.393^{***}	2.034^{***}	1.395^{***}	1.778^{**}
	(.600)	(.757)	(.528)	(.717)	(.383)	(.707)
Ownership^2	-1.521^{*}	-1.377	-2.422^{***}	-3.610^{***}	-2.851^{***}	-2.699^{***}
	(.827)	(.991)	(.683)	(1.174)	(.695)	(.957)
Observations	23,879	23,766	21,516	23,879	23,879	23,879

- OLS, FD, PRL, and DML show a positive level term and negative curvature.
- Both FE terms are individually insignificant.
- Marginal effects depend on the initial ownership level.

Current manuscript, Table 5. Firm-clustered SE; full controls. FE excludes 113 singletons. * $p < .10$, ** $p < .05$, *** $p < .01$.

Robustness and interpretation

R&D application

- Alternative innovation outcomes and R&D measures.
- Patenting-firm sample and observations with reported R&D.
- Alternative tuning choices and cross-fitting folds.

Ownership application

- Top-five-executive ownership and alternative sample periods.
- Results vary across periods; historical-sample curvature is not uniformly concave.

These checks assess sensitivity. They do not establish a unique causal explanation for differences across estimators.

Current manuscript, Sections 5.2–5.3 and the corresponding Internet Appendix robustness tables.

Simulation design

The count outcome follows a negative binomial distribution:

$$\mathbb{E}[y_{it} \mid z_{it}, x_{it}] = \exp(0.2 z_{it} + x_{it}).$$

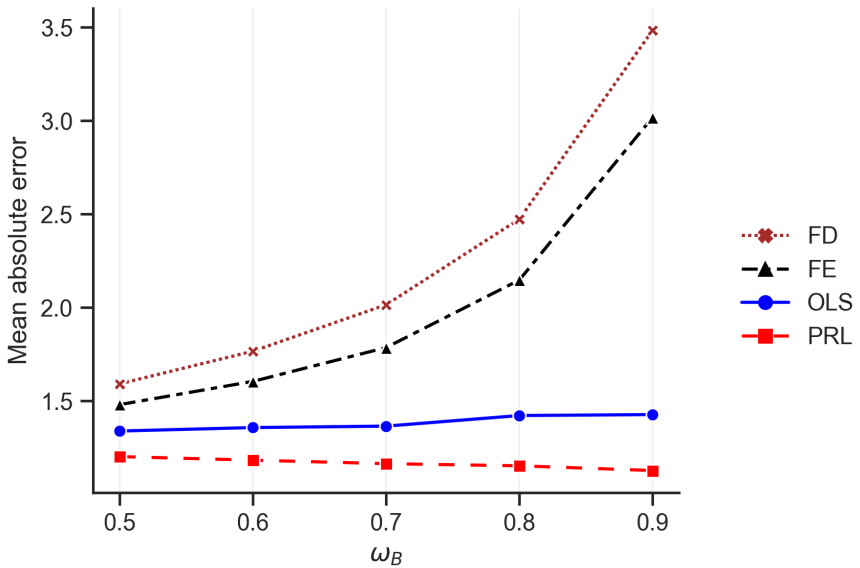
- $N = 1{,}000$, $T = 10$, dispersion parameter 0.5; 10,000 replications per cell.
- Four designs: between-firm share, serial persistence, prevalence of a firm component, and a component shared with an observed control.
- Linear comparisons: OLS, FE, FD, PRL. Count comparisons: Poisson, FE Poisson, PRL Poisson.
- MSE and MAE use the paper's common reference $\gamma = 0.2$; one-sided power is $\Pr(t > 1.645)$.
- DML is omitted from simulations because of computational cost.

Current manuscript, Section 3. Linear estimators use the same count outcomes; the exponential-mean coefficient is their common comparison benchmark, not an assumed correctly specified linear slope.

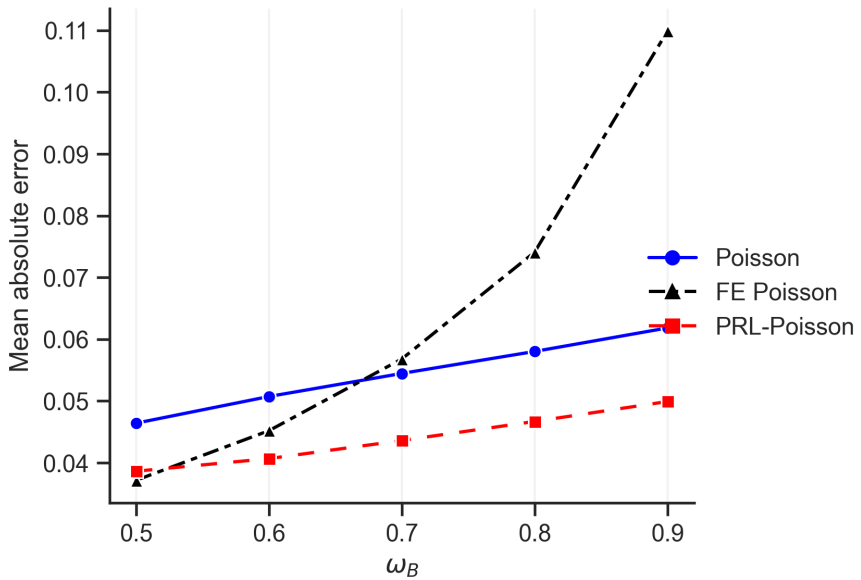
Between-firm variation

$z_{it} = 0.5\eta_i + 0.5\nu_{it}$; horizontal axis: ω_B.

Linear: MAE



Poisson: MAE



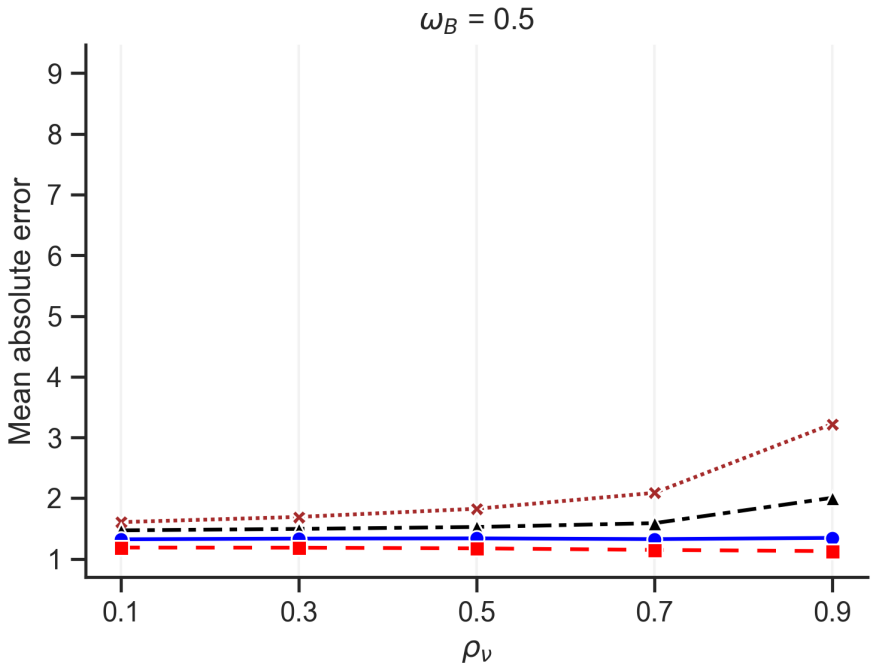
- Linear PRL has the lowest error in these cells.
- FE Poisson performs well at moderate between-firm shares; its error rises when the share becomes high.

Current manuscript, Figure 1, MAE panels. Original curves and scales retained; $N = 1,000$, $T = 10$, 10,000 replications.

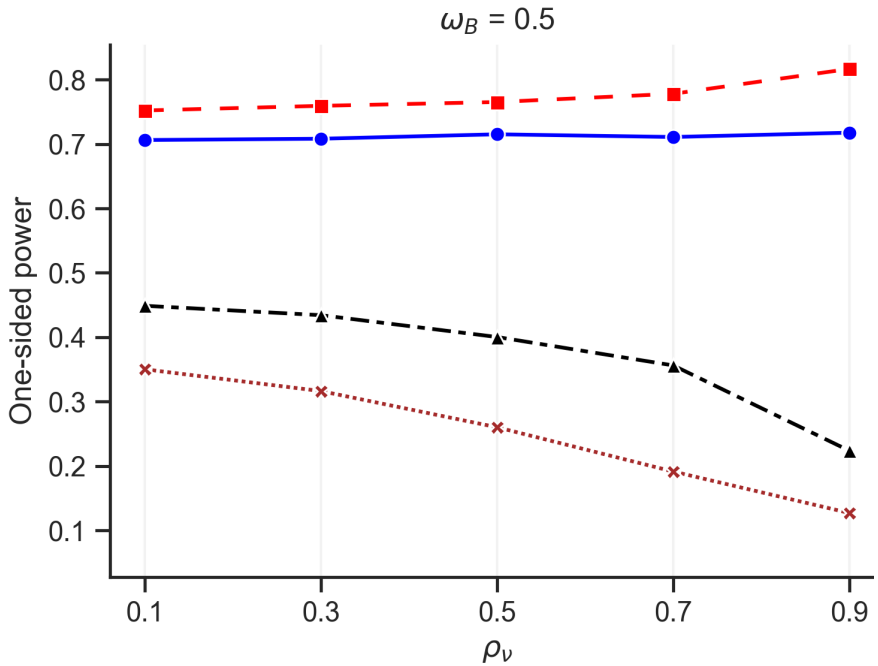
Serial persistence

$$\nu_{it} = \rho_\nu \nu_{i,t-1} + \sigma_\nu \sqrt{1 - \rho_\nu^2}\, \varepsilon_{\nu,it}.$$

Mean absolute error



One-sided power



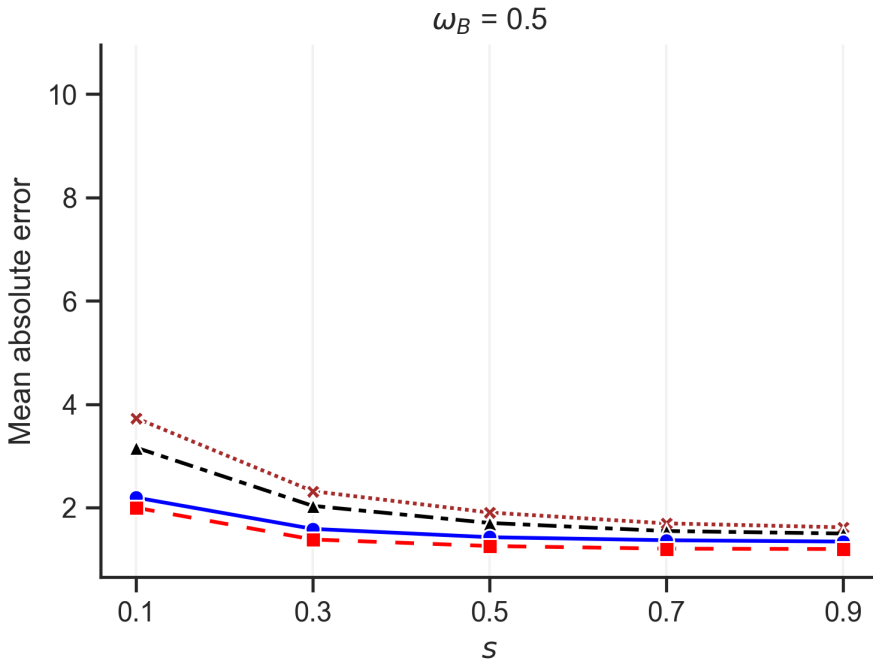
OLS: blue solid; PRL: red dashed; FE: black dash-dot; FD: brown dotted.

FE and FD errors increase and power declines as persistence rises, particularly when the between-firm share is high.

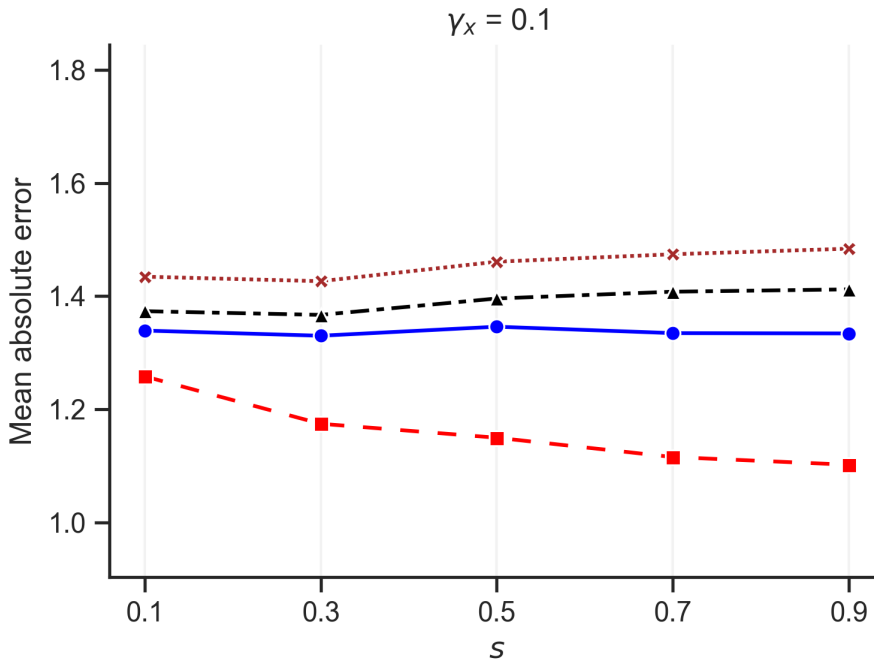
Current manuscript, Figure 2. Enlarged $\omega_B = 0.5$ cells; the paper also reports 0.7 and 0.9. Original curves and scales retained.

Prevalence and shared firm components

Prevalence: linear MAE



Shared component: linear MAE



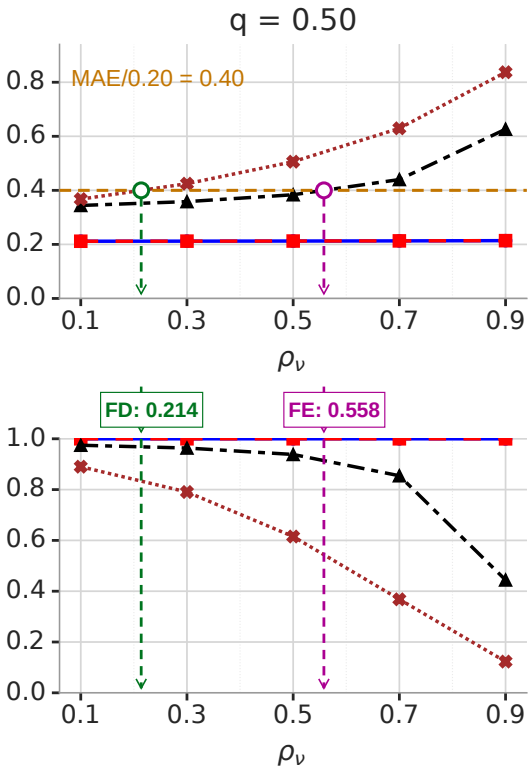
OLS: blue solid; PRL: red dashed; FE: black dash-dot; FD: brown dotted.

PRL performs well in the linear comparisons. Poisson rankings also depend on the design and error criterion.

Current manuscript, Figures 3 and 4. Enlarged MAE cells: $\omega_B = 0.5$ for prevalence and $\gamma_x = 0.1$ for the shared component. Backup slides show additional parameter cells.

Diagnostic guidance under an assumed DGP

$z = z^* + m$, $y = \alpha_i + 0.2z^* + \varepsilon$, $\mathrm{Var}(z^*) = 1$.
Calibration: $\sigma_\varepsilon = 3$, $\omega_B = 0.5$, $q = \sigma_m^2/\sigma_\nu^2 = 0.5$.



OLS: blue; PRL: red; FE: black; FD: brown.

Target: MAE/0.2 $<$ 0.4

- FD crosses at $\rho_\nu = 0.214$.
- FE crosses at $\rho_\nu = 0.558$.

Target: power $\geq$ 0.9

- FD fails throughout the plotted range.
- FE crosses near $\rho_\nu = 0.591$.

The diagnostics translate these crossings into observable benchmarks *under this calibration*.

Current manuscript, Section 4.3 and Figure 5; IA power guide. Cropped middle column of the original MAE guide. Cutoffs are interpolated and scenario-specific.

Adjusted Hausman–Taylor: supplementary evidence

- The adjusted HT estimator uses internal instruments and GMM moments involving demeaned regressors.
- Its validity requires strict exogeneity and additional restrictions on controls used as instruments in levels.
- R&D coefficients: patents 0.220, citations 0.551, and adjusted citations 0.231; all significant at 1%.
- Ownership: level 1.778 and square -2.699, significant at 5% and 1%, respectively.
- We treat it as an assumption-dependent supplementary comparison.

Current manuscript, Section 6 and Tables 2 and 5; Hausman and Taylor (1981).

Stata implementation

The linear application uses BIC and five cross-fitting folds:

```
poregress y z x, controls(i.firmID i.year) ///
    vce(cluster firmID) selection(bic, gridminok) ///
    rseed(1103)

xporegress y z x, controls(i.firmID i.year) ///
    vce(cluster firmID) selection(bic, gridminok) ///
    xfolds(5) rseed(1103)
```

- `x` stands for the full control-variable list.
- Poisson counterparts: `popoisson` and `xpopoisson`.
- The sparse package supplies `sparse_poreg` and `sparse_xporeg` for large *linear* models.

Commands adapted from `RDPatent/po_xpo_adjHT.do`. Replication code, example data, and sparse Stata package.

Practical reporting guidance

1. Report between-firm variation, within-firm persistence, FE absorption, and between/within correlation R^2.
2. Explain the assumptions supporting pooled, FE, FD, and selective specifications.
3. Report estimation observations, singleton removal, and the annual-pair sample for FD.
4. Report selected firm/year counts and sensitivity to tuning and folds.
5. Use a substantively justified DGP when translating diagnostics into error and power targets.

Current manuscript, Section 4. Diagnostics characterize the available variation and do not independently validate identification.

Conclusions

- Persistent regressors create a difficult balance between controlling heterogeneity and preserving useful variation.
- PRL and DML provide selective specifications for linear and Poisson applications.
- Simulations identify settings where PRL improves performance, while also showing that no estimator dominates uniformly.
- The two applications yield positive R&D–innovation associations and concave ownership–value relations under selective estimation.
- Interpretation requires attention to identifying assumptions, measurement quality, and available within-firm information.

Current manuscript, Conclusion. Empirical coefficient differences alone do not establish causality or a unique explanation for the differences.

Thank you!

Questions and comments

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Paper: SSRN 4636846

Replication code, example data, and sparse Stata package

Estimating Persistent Explanatory Variables:
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Diagnostic definitions

D1: share of regressor variation between firms

$$\widehat{\omega}_B = \frac{\sum_i T_i(\bar{z}_i - \bar{z})^2}{\sum_i T_i(\bar{z}_i - \bar{z})^2 + \sum_{it}(z_{it} - \bar{z}_i)^2}.$$

D2: within-firm persistence

$$\widehat{\rho}_i = \mathrm{Corr}_t(z_{it}, z_{i,t-1}).$$

Use consecutive years; report the median and interquartile range across firms.

D3: absorption by the specified FE structure

$$A_z = 1 - \frac{\sum_{it} \widehat{v}_{z,it}^2}{\sum_{it}(z_{it} - \bar{z})^2}.$$

$\widehat{v}_{z,it}$ is the residual from regressing z on those effects. With firm effects only, D3 equals D1.

Current manuscript, Section 4.1. T_i is firm i's observation count; bars denote sample means.

Between- and within-firm R-squared

Use predictions from observed regressors and year effects:

$$\widehat{m}_{i,t+1} = \widehat{\beta}_0 + \widehat{\gamma} z_{it} + \mathbf{x}_{it}^{\top} \widehat{\boldsymbol{\beta}}.$$

These predictions exclude the estimated firm indicators.

$$R^2_{\mathrm{B,corr}} = \mathrm{Corr}_i(\bar{y}_i, \bar{m}_i)^2,$$

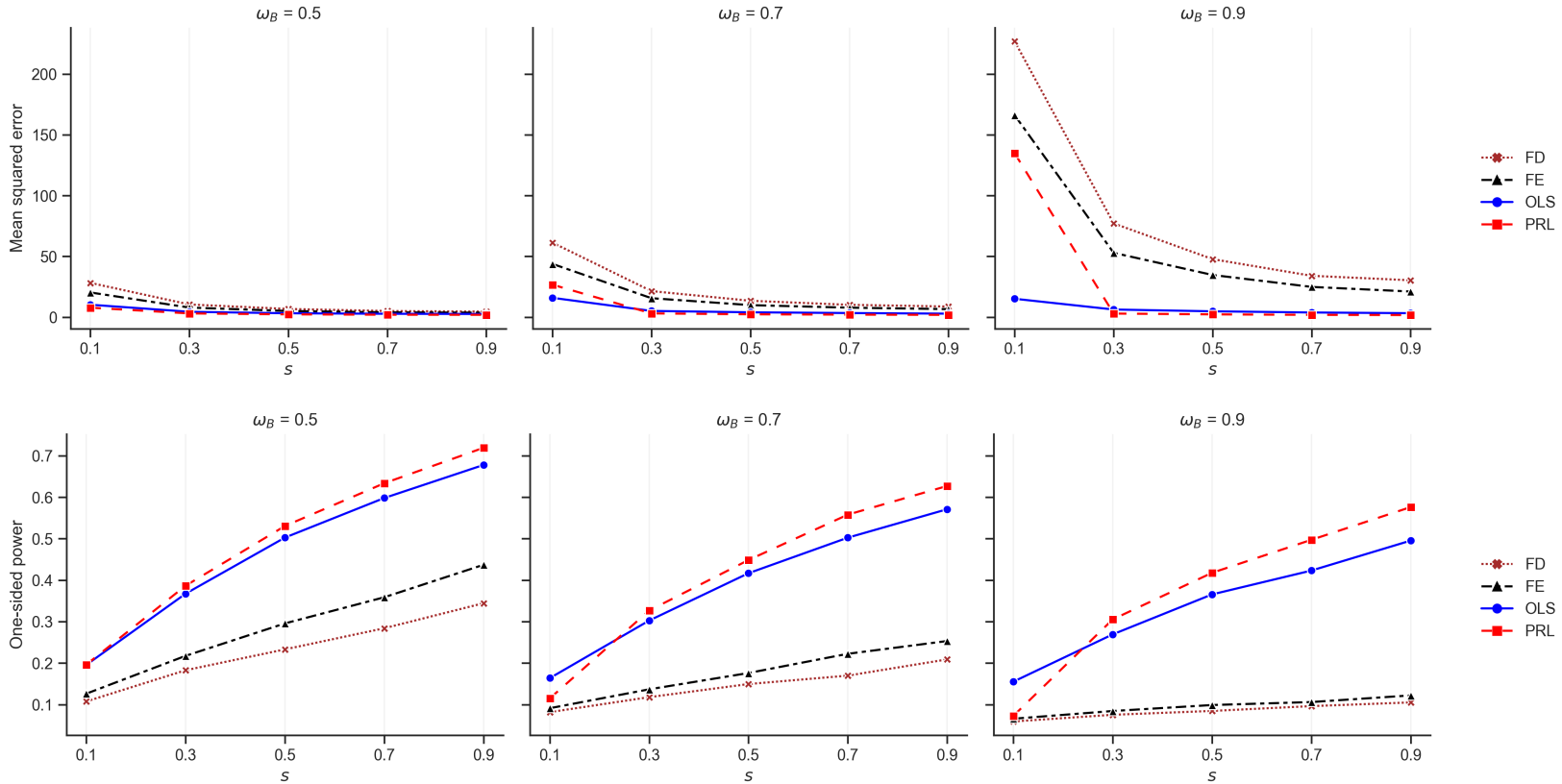
$$R^2_{\mathrm{W,corr}} = \mathrm{Corr}_{it}(y_{i,t+1} - \bar{y}_i, \widehat{m}_{i,t+1} - \bar{m}_i)^2.$$

- The between measure compares firm-average outcomes and predictions.
- The within measure compares deviations from firm means.
- These differ from full-model R^2 including fitted firm effects, and generally depend on the fitted slopes.

Current manuscript, Section 4.1. These squared correlations are not additive variance shares.

Prevalence of the time-invariant component

$z_{it} = 0.5 D_i \eta_i + 0.5 \nu_{it}$, $D_i \sim$ Bernoulli(s).

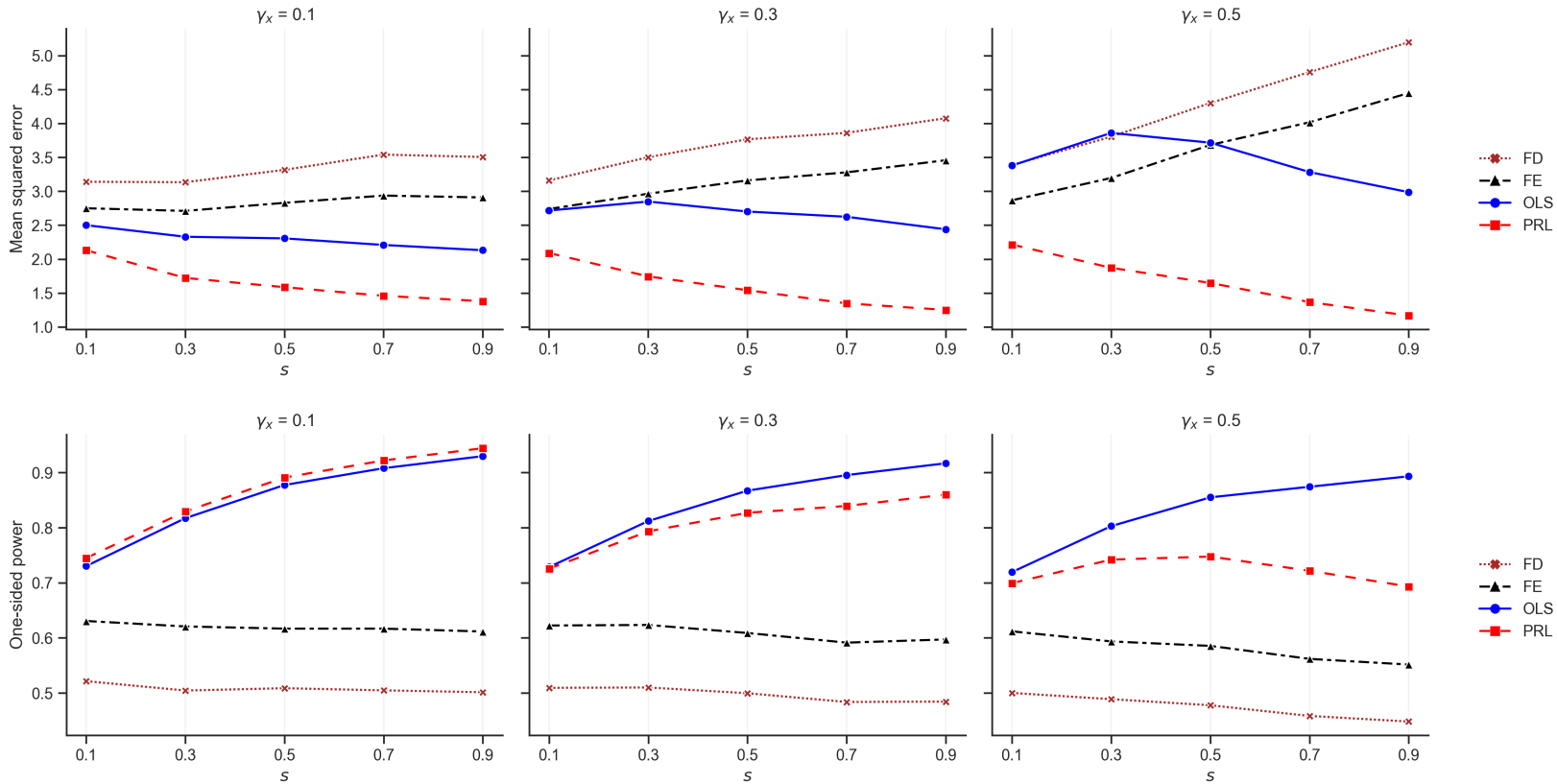


OLS: blue solid; PRL: red dashed; FE: black dash-dot; FD: brown dotted.

Current manuscript, Figure 3. Rows: MSE and power. Columns: ω_B = .5, .7, .9. The scale $\sigma_\nu^2 = s\sigma_\eta^2(1-\omega_B)/\omega_B$ holds the population share fixed as s varies.

A component shared by the regressor and control

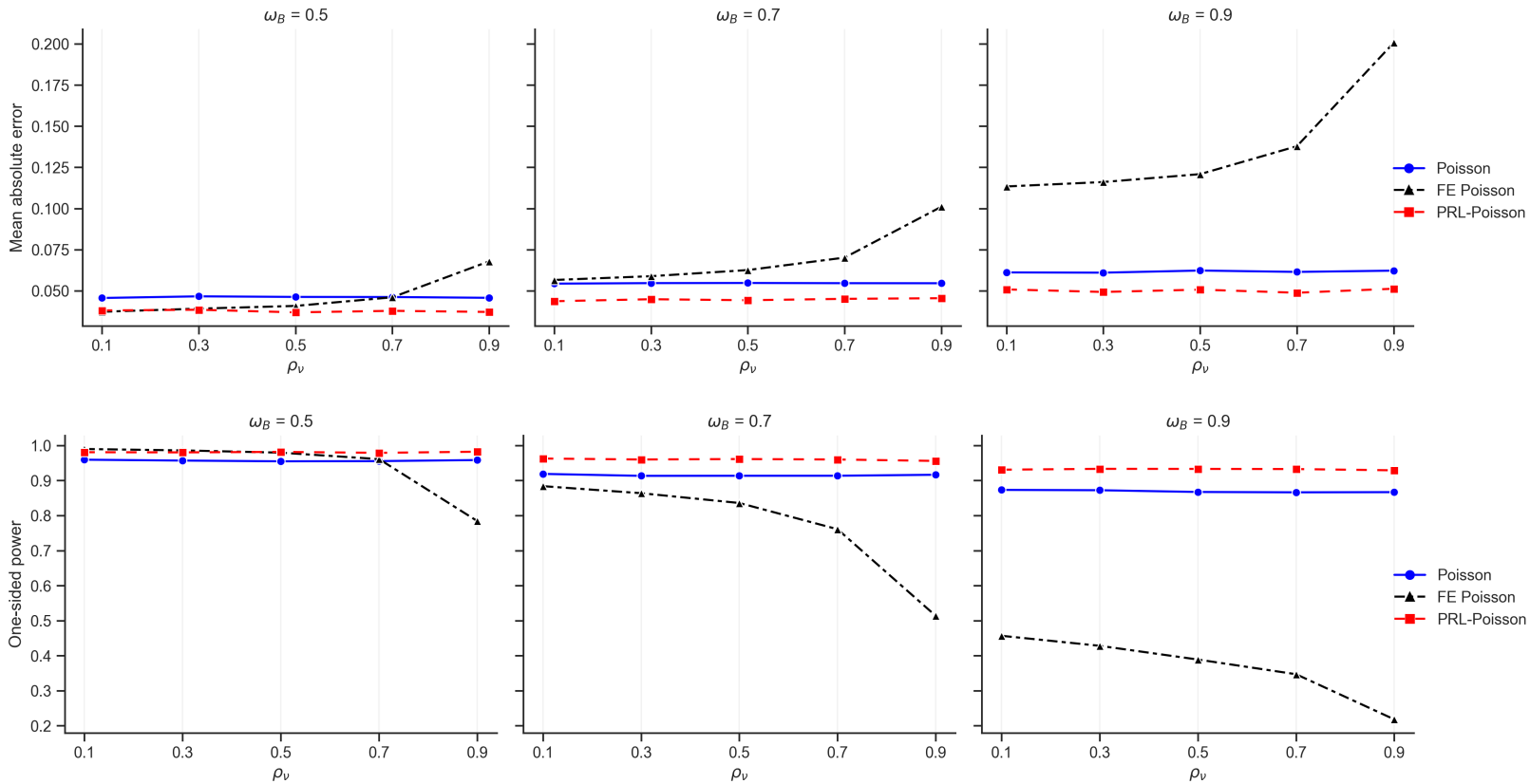
$$z_{it} = D_i\eta_i + \nu_{it},\ x_{it} = \gamma_x D_i\eta_i + \kappa_{it}.$$



OLS: blue solid; PRL: red dashed; FE: black dash-dot; FD: brown dotted.

Current manuscript, Figure 4. Rows: MSE and power. Columns: $\gamma_x = .1, .3, .5$. Baseline variance calibration .7; the population between-firm variance share is $0.7s/(0.7s + 0.3)$.

Serial persistence: Poisson comparisons



FE Poisson can perform well with moderate between-firm variation and low persistence. Its performance deteriorates in more persistent settings; PRL Poisson does not dominate every cell.

Internet Appendix, count-outcome serial-persistence figure. Rows: MAE and power. Columns: ω_B = .5, .7, .9; 10,000 replications.

Larger panels and continuous outcomes

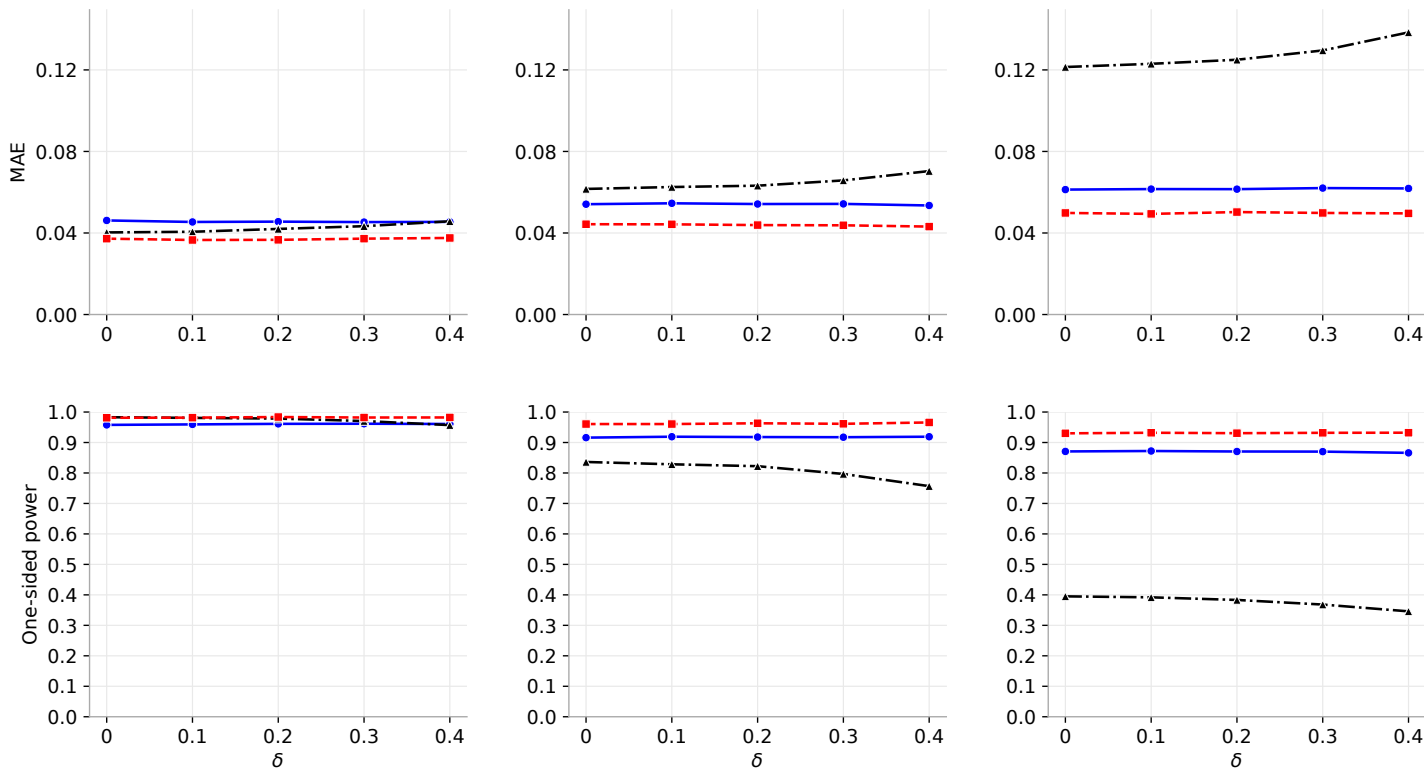
- Larger panels: $N = 5{,}000$, $T = 10$, and 2,000 replications per parameter combination.
- Continuous outcome: a nonnegative lognormal–chi-squared mixture with conditional mean $\exp(0.2z_{it} + x_{it})$.
- The Internet Appendix repeats the between-firm, serial-persistence, prevalence, and shared-component designs.
- Larger panels generally reduce errors and increase power, but relative rankings depend on the design and loss criterion.
- The sparse linear PRL package reduces memory and computational costs in large panels.

Current manuscript, Section 3 and Internet Appendix simulation extensions. These simulation comparisons omit DML.

Heterogeneous persistence across firms

Half the firms have $\rho_L = 0.5 - \delta$ and half $\rho_H = 0.5 + \delta$.

Columns: $\omega_B = 0.5, 0.7, 0.9$. Rows: MAE and one-sided power.



Poisson: blue solid; FE Poisson: black dash-dot; PRL Poisson: red dashed.

Internet Appendix, count outcome, $N = 1,000$, $T = 10$, 10,000 replications. $\delta = 0, .1, .2, .3, .4$. PRL Poisson has the lowest MAE throughout.

R&D robustness: patenting firms

Dependent variable: ln(1 + Patents)

	OLS	FE	FD	PRL	DML	adjHT
R&D/Assets	.448***	.035	−.003	.406***	.399***	.161***
	(.045)	(.031)	(.028)	(.044)	(.025)	(.030)
Selected firms				128	255	
Firm selection				2.97%	5.91%	
Observations	45,913	45,679	40,944	45,913	45,913	45,913

- Restrict to firms with at least one patent during the sample period.
- The positive pooled and selective associations remain.
- Selected proportions are lower than in the full sample.

Internet Appendix, table labelled `tab:rd_linear_patenting`, Panel A. Full controls; firm-clustered SE. FE excludes 234 singletons. *** $p < .01$.

Ownership robustness: top five executives

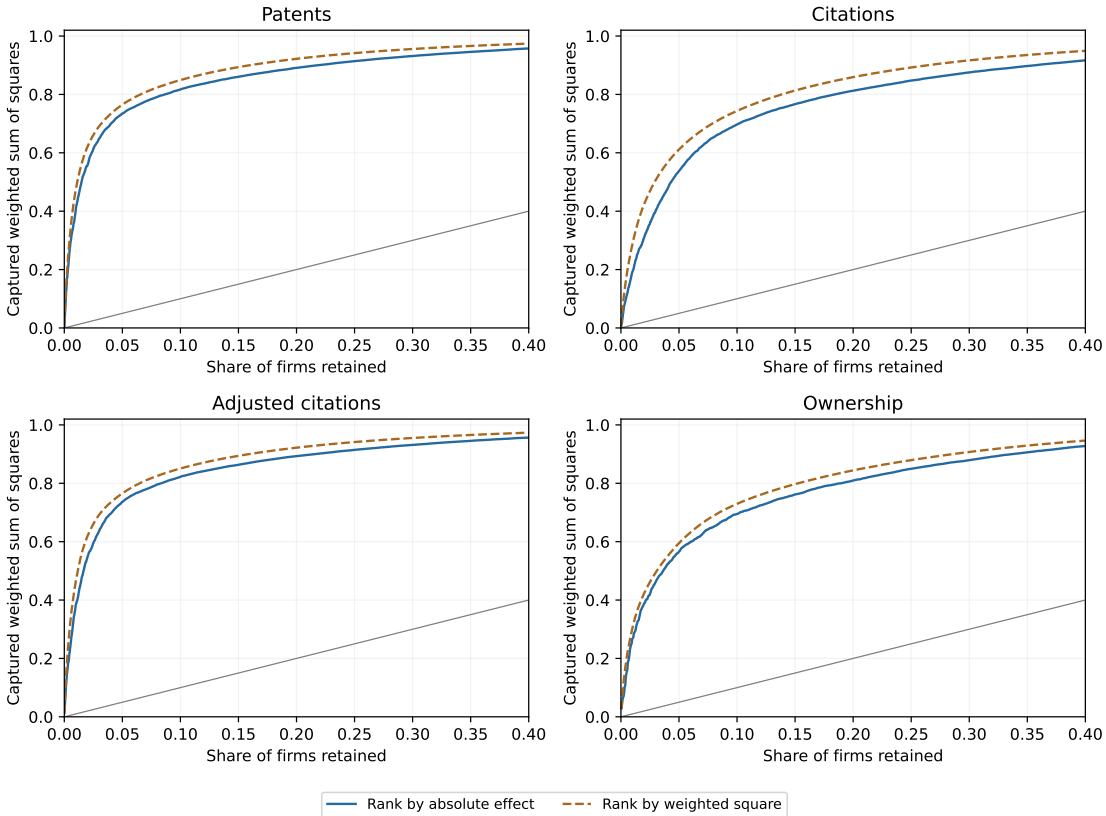
Dependent variable: Tobin's Q; level sample 2007–2024.

	OLS	FE	FD	PRL	DML	adjHT
Ownership	2.066^{***}	$-.035$	1.244^{**}	1.305^{**}	1.201^{***}	2.017^{***}
	(.608)	(.754)	(.503)	(.524)	(.402)	(.711)
Ownership^2	-1.612^{*}	-1.399	-2.183^{***}	-2.326^{***}	-2.357^{***}	-2.704^{***}
	(.832)	(.991)	(.687)	(.860)	(.733)	(.964)
Observations	23,892	23,779	21,522	23,892	23,892	23,892

The positive level and negative squared terms remain under pooled and selective estimation. FE terms remain individually insignificant.

Internet Appendix, `ownership_top5_main_period_table.tex`. Scope-specific filters yield a slightly different sample from the all-named-executive baseline. Firm-clustered SE; conventional significance stars.

Concentration of estimated firm effects



A limited subset can capture a large share of the fitted firm-effect component in some applications. This is descriptive evidence, affected by estimation noise.

Internet Appendix concentration analysis. Solid: rank by absolute centered effect; dashed: rank by observation-weighted squared effect. These are unrestricted-FE rankings, not actual PRL/DML selected sets.

Implementation and selected references

- Chernozhukov, Hansen, and Spindler (2015): post-regularization inference.
- Chernozhukov et al. (2018): double/debiased machine learning.
- Belloni, Chernozhukov, and Wei (2016): post-selection inference for generalized linear models.
- Breuer and deHaan (2024): use and interpretation of fixed effects.
- Jennings et al. (2024): measurement error and fixed effects.
- Cohn, Liu, and Wardlaw (2022): count-data regression in finance.

Code, data, logs, and the sparse linear package

Replication code, example data, and sparse Stata package

Paper: SSRN 4636846

Slide content follows the local revised manuscript, `Persistent_FE_v1007.tex`, read 31 October 2026. Full bibliographic details are in the paper.